



**Teheran University of Medical Sciences**

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**Final Exam**  
**International Campus, MD Program**  
**Cardiovascular System**  
**September 2022 semester**

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Course Coordinator: Dr. Ranjbaran

Date of Exam: Jan 04, 2023

Duration of Exam: 100 Minutes

MCQ Question

100

**Directions:**

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.



Physiology: Dr. Ranjbaran

|                                                                                                           |     |
|-----------------------------------------------------------------------------------------------------------|-----|
| Blood Physiology                                                                                          |     |
| A person who is suffering from chronic kidney disease may need to receive ..... in order to produce RBCs. | 1   |
| erythropoietin                                                                                            | الف |
| whole blood                                                                                               | ب   |
| bone marrow transplantation                                                                               | ج   |
| fresh frozen plasma                                                                                       | د   |
| منبع:                                                                                                     |     |

|                                                                                 |     |
|---------------------------------------------------------------------------------|-----|
| Blood Physiology                                                                |     |
| Which of the following is considered to be the main cause of "Aplastic Anemia"? | 2   |
| Lack of functioning bone marrow                                                 | الف |
| Lack of RBC function                                                            | ب   |
| Lack of carbonic anhydrase enzyme                                               | ج   |
| Iron deficiency                                                                 | د   |
| منبع:                                                                           |     |

|                                                                                                   |     |
|---------------------------------------------------------------------------------------------------|-----|
| Blood Physiology                                                                                  |     |
| Which of the following is <u>not considered</u> to be one of the characteristics of platelets?    | 3   |
| They are minute discs 1 to 4 micrometers in diameter.                                             | الف |
| They are formed in the bone marrow from megakaryocytes.                                           | ب   |
| The normal concentration of platelets in the blood is between 150,000 and 300,000 per microliter. | ج   |
| They have nuclei and can reproduce.                                                               | د   |
| منبع:                                                                                             |     |

|                                                                                     |     |
|-------------------------------------------------------------------------------------|-----|
| Blood Physiology                                                                    |     |
| Which of the following is known as a white blood cell with the stronger chemotaxis? | 4   |
| Neutrophils                                                                         | الف |
| Basophils                                                                           | ب   |
| Eosinophils                                                                         | ج   |
| Lymphocytes                                                                         | د   |
| منبع:                                                                               |     |

|                                                                               |     |   |
|-------------------------------------------------------------------------------|-----|---|
| <b>Blood Physiology</b>                                                       |     |   |
| Which of the following refers to the main importance of "Walling off" effect? |     | 5 |
| To accelerate chemicals release                                               | الف |   |
| To decrease cytokines secretion                                               | ب   |   |
| To accelerate inflammatory responses                                          | ج   |   |
| To delay the spread of bacteria                                               | د   |   |
| منبع:                                                                         |     |   |

|                                                                               |     |   |
|-------------------------------------------------------------------------------|-----|---|
| <b>Blood Physiology</b>                                                       |     |   |
| Which mediators are involved in the second line of defense against infection? |     | 6 |
| Selectin and ICAM-1                                                           | الف |   |
| GM-CSF and G-CSF                                                              | ب   |   |
| Living bacteria and cellular exudate                                          | ج   |   |
| Histamin and bradikinine                                                      | د   |   |
| منبع:                                                                         |     |   |

|                                                                                      |     |   |
|--------------------------------------------------------------------------------------|-----|---|
| <b>Blood Physiology</b>                                                              |     |   |
| Immediately after a blood vessel is ruptured, trauma to the vessel wall causes ..... |     | 7 |
| vascular constriction                                                                | الف |   |
| platelet plug formation                                                              | ب   |   |
| intrinsic pathway of blood coagulation activation                                    | ج   |   |
| extrinsic pathway of blood coagulation activation                                    | د   |   |
| منبع:                                                                                |     |   |

|                                                                          |     |   |
|--------------------------------------------------------------------------|-----|---|
| <b>Blood Physiology</b>                                                  |     |   |
| In the opposite figure, what should be written instead of question mark? |     | 8 |
|                                                                          |     |   |
| Factor X                                                                 | الف |   |
| Vascular trauma                                                          | ب   |   |
| Prothrombin activator                                                    | ج   |   |
| Factor XII                                                               | د   |   |
| منبع:                                                                    |     |   |



Blood Physiology

Which of the following is not included in the fibrinolytic system?

Plasminogen

Tissue plasminogen activator

plasminogen activator inhibitor

Protein C

الف

ب

ج

د

منبع:

Blood Physiology

Which organ produces the largest number of red blood cells in the middle trimester of pregnancy?

10

Liver

Spleen

Bone marrow

Lymph node

الف

ب

ج

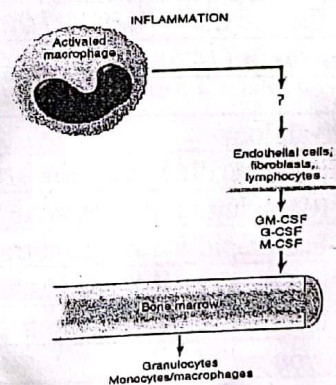
د

منبع:

Blood Physiology

In control of bone marrow production of granulocytes and monocyte-macrophages, what should be written instead of question mark?

11



TNF

IL-6

Selectin

ICAM-1

الف

ب

ج

د

منبع:

|                                                                    |     |       |
|--------------------------------------------------------------------|-----|-------|
| <b>Blood Physiology</b>                                            |     |       |
| The least amount of iron in the body is found in the form of ..... |     | 12    |
| Transferrin                                                        | الف |       |
| Ferritin                                                           | ب   |       |
| Myoglobin                                                          | ج   |       |
| Hemoglobin                                                         | د   |       |
|                                                                    |     | منبع: |

**Physiology: Dr. Sohanaki**

|                                              |     |       |
|----------------------------------------------|-----|-------|
| <b>Physiology</b>                            |     |       |
| <b>Atrial systole</b>                        |     | 13    |
| a- Causes a decrease in atrial pressure      | الف |       |
| b- Causes the 'a' wave of the jugular pulse  | ب   |       |
| c- Causes the 'c' wave of the jugular pulse  | ج   |       |
| d- Causes the 'v' wave of the jugular pulse. | د   |       |
| Guyton p.114                                 |     | منبع: |

|                                                                            |     |       |
|----------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                          |     |       |
| <b>During cardiac cycle the atrial pressure:</b>                           |     | 14    |
| a- Rises during atrial systole and drops during isovolumetric contraction. | الف |       |
| b- Falls rapidly during ventricular ejection.                              | ب   |       |
| c- Rises as blood flows passively into the ventricles.                     | ج   |       |
| d- Decreased markedly in tricuspid regurgitation.                          | د   |       |
| Guyton p.114                                                               |     | منبع: |

|                                                                                   |     |       |
|-----------------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                                 |     |       |
| <b>During isovolumetric ventricular contraction phase</b>                         |     | 15    |
| a- Mitral and tricuspid valves closure causing second heart sound                 | الف |       |
| b- Intraventricular pressure is increased without change in ventricular volume.   | ب   |       |
| c- Atrioventricular valves bulge into the atria causing a drop in atrial pressure | ج   |       |
| d- QRS complex coincides with ventricular contraction                             | د   |       |
| Guyton p.115                                                                      |     | منبع: |



|                                                                                            |     |       |
|--------------------------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                                          |     |       |
| <b>Regarding atrial systole all the followings occurs EXCEPT:</b>                          |     |       |
| a- The mitral and tricuspid valves are open and the aortic and pulmonary valves are closed | الف |       |
| b- Contraction of the atria pushes 30% of blood into the ventricles.                       | ب   |       |
| c- Contraction of the atrial muscle prevents regurgitation of blood into the veins.        | ج   |       |
| d- P wave precedes atrial systole.                                                         | د   |       |
| Guyton p.114                                                                               |     | منبع: |

|                                                                                                           |     |        |
|-----------------------------------------------------------------------------------------------------------|-----|--------|
| <b>Physiology</b>                                                                                         |     |        |
| <b>Starling's law of the heart</b>                                                                        |     | 1<br>7 |
| a. concerns the relationship between the degree of stretch of cardiac muscle and the force of contraction | الف |        |
| b. states that 'the stroke volumes of the left and right ventricles are matched                           | ب   |        |
| c. causes an increase of contractility of cardiac muscle                                                  | ج   |        |
| d. can equally be applied to skeletal as well as cardiac muscle                                           | د   |        |
| Guyton p.119                                                                                              |     | منبع:  |

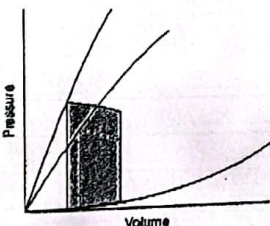
|                                                                                                            |     |        |
|------------------------------------------------------------------------------------------------------------|-----|--------|
| <b>Physiology</b>                                                                                          |     |        |
| <b>During ventricular ejection:</b>                                                                        |     | 1<br>8 |
| a- The left ventricular pressure is increased to 120 mmHg and that of the right ventricle to 80 mmHg.      | الف |        |
| b- Late in systole the momentum keeps the blood moving forward because the aortic pressure drops suddenly. | ب   |        |
| c- The amount of the blood ejected by each ventricle per cycle at rest is 130 mL.                          | ج   |        |
| d- The atrioventricular valves are pulled down and atrial pressure drops.                                  | د   |        |
| Guyton p.115                                                                                               |     | منبع:  |

|                                                         |     |        |
|---------------------------------------------------------|-----|--------|
| <b>Physiology</b>                                       |     |        |
| <b>Funny currents persist...</b>                        |     | 1<br>9 |
| a. at all times throughout SA Nodal cell AP propagation | الف |        |
| b. only when an SA nodal cell is hyperpolarized         | ب   |        |
| c. only during the repolarization of an SA nodal cell   | ج   |        |
| d. none of the above.                                   | د   |        |
| Guyton p.124                                            |     | منبع:  |

|                                                                                                                          |     |       |
|--------------------------------------------------------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                                                                        |     | 20    |
| <b>The ionic basis of the automaticity in the nodal cells that mediates the slow diastolic depolarization is all but</b> |     |       |
| a. hyperpolarization-induced $\text{Na}^+$ inward current ( $i_f$ )                                                      | الف |       |
| b. an outward $i_K$ current ( $i_K$ )                                                                                    | ب   |       |
| c. an inward rectifying $\text{K}^+$ current ( $i_{K1}$ )                                                                | ج   |       |
| d. an inward $\text{Ca}^{2+}$ current through L-type $\text{Ca}^{++}$ channels ( $i_{Ca}$ )                              | د   |       |
| Guyton p.124                                                                                                             |     | منبع: |

|                                                                                                       |     |       |
|-------------------------------------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                                                     |     | 21    |
| <b>In aortic pressure curve:</b>                                                                      |     |       |
| a- Ventricular systole stretches aortic and arterial walls and causes systolic pressure of 120 mm Hg. | الف |       |
| b- Dicrotic notch occurs when the aortic valve opens.                                                 | ب   |       |
| c- Incisura is caused by forward flow of blood before closure of the valve.                           | ج   |       |
| d- Elastic recoil of the aorta and arterial walls maintain diastolic pressure of 80 mm Hg.            | د   |       |
| Guyton p: 116                                                                                         |     | منبع: |

|                                                                                      |     |       |
|--------------------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                                    |     | 22    |
| <b>Cardiac sympathetic nerve activation:</b>                                         |     |       |
| a- Shifts the cardiac output curve downwards and to the right.                       | الف |       |
| b- Increases the formation of cAMP via $\beta_1$ receptors.                          | ب   |       |
| c- Deactivates the ligand-gated $\text{Ca}^{2+}$ channels of the ryanodine receptor. | ج   |       |
| d- Decreases active reuptake of $\text{Ca}^{2+}$                                     | د   |       |
| Guyton p.120                                                                         |     | منبع: |

|                                                                                                                                                                                        |     |    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| <b>Physiology</b>                                                                                                                                                                      |     | 23 |
| <b>Below graph shows the pressure- volume curve of the left ventricle of a normal subject (broken lines) and a patient. Compared to normal, which term best describes this patient</b> |     |    |
|                                                                                                     |     |    |
| a. Decreased afterload                                                                                                                                                                 | الف |    |
| b. Decreased preload                                                                                                                                                                   | ب   |    |
| c. Decreased stroke volume                                                                                                                                                             | ج   |    |
| d. Increased force of contraction                                                                                                                                                      | د   |    |



## Physiology

Left axis deviation of the ECG is described when:

- |                                                                                |     |
|--------------------------------------------------------------------------------|-----|
| a. the axis of the heart is from $-30^{\circ}$ to $+90^{\circ}$ .              | الف |
| b. the QRS complex is prominent positive in lead aVF.                          | ب   |
| c. the QRS is prominent positive in lead I and prominent negative in lead III. | ج   |
| d. right ventricle is enlarged                                                 | د   |

Guyton p.145

منبع:

## Physiology

Regarding the conduction system of the heart

25

- |                                                                                    |     |
|------------------------------------------------------------------------------------|-----|
| a. the right bundle branch divides into anterior and posterior fascicles           | الف |
| b. the AV node contains P and T cells                                              | ب   |
| c. myocardial fibers have a resting membrane potential of $-60$ mV                 | ج   |
| d. action potential in the SA and AV nodes are largely due to $\text{Na}^+$ influx | د   |

Guyton p: 124

منبع:

## Physiology

Which of the following changes will cause an increase in myocardial  $\text{O}_2$  consumption?

26

- |                                                                                |     |
|--------------------------------------------------------------------------------|-----|
| a. Decreased aortic pressure                                                   | الف |
| b. Decreased heart rate                                                        | ب   |
| c. Increased size of the heart                                                 | ج   |
| d. Increased influx of $\text{Na}$ during the upstroke of the action potential | د   |

Guyton p: 119

منبع:

## Physiology

Considering conduction rates in myocardial cells, which statement is TRUE

27

- |                                                                                                      |     |
|------------------------------------------------------------------------------------------------------|-----|
| a. Purkinje fibers are subepicardial and the largest fibers, 4-7 times the width of other fibers     | الف |
| b. Purkinje fibers are 'fast fibers', and can conduct a wave of depolarization at a speed of 4m/sec  | ب   |
| c. Duration of the action potential and refractory period in fast fibers is shorter than slow fibers | ج   |
| d. Initial depolarization occurs in fast fibers with a rapid influx of $\text{Ca}^{2+}$ ions         | د   |

Guyton p 126

منبع:



|                                                                                |     |
|--------------------------------------------------------------------------------|-----|
| <b>Physiology</b>                                                              |     |
| <b>Left axis deviation of the ECG is described when:</b>                       |     |
| a. the axis of the heart is from $-30^{\circ}$ to $+90^{\circ}$ .              | الف |
| b. the QRS complex is prominent positive in lead aVF.                          | ب   |
| c. the QRS is prominent positive in lead I and prominent negative in lead III. | ج   |
| d. right ventricle is enlarged                                                 | د   |
| Guyton p.145                                                                   |     |
| منبع:                                                                          |     |

|                                                                                      |     |       |
|--------------------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                                    |     |       |
| <b>Regarding the conduction system of the heart</b>                                  |     | 25    |
| a. the right bundle branch divides into anterior and posterior fascicles             | الف |       |
| b. the AV node contains P and T cells                                                | ب   |       |
| c. myocardial fibers have a resting membrane potential of $-60$ mV                   | ج   |       |
| d. action potential in the SA and AV nodes are largely due to $\text{Na}^{+}$ influx | د   |       |
| Guyton p: 124                                                                        |     | منبع: |

|                                                                                                                 |     |       |
|-----------------------------------------------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                                                               |     |       |
| <b>Which of the following changes will cause an increase in myocardial <math>\text{O}_2</math> consumption?</b> |     | 26    |
| a. Decreased aortic pressure                                                                                    | الف |       |
| b. Decreased heart rate                                                                                         | ب   |       |
| c. Increased size of the heart                                                                                  | ج   |       |
| d. Increased influx of Na during the upstroke of the action potential                                           | د   |       |
| Guyton p: 119                                                                                                   |     | منبع: |

|                                                                                                      |     |       |
|------------------------------------------------------------------------------------------------------|-----|-------|
| <b>Physiology</b>                                                                                    |     |       |
| <b>Considering conduction rates in myocardial cells, which statement is TRUE</b>                     |     | 27    |
| a. Purkinje fibers are subepicardial and the largest fibers, 4-7 times the width of other fibers     | الف |       |
| b. Purkinje fibers are 'fast fibers', and can conduct a wave of depolarization at a speed of 4m/sec  | ب   |       |
| c. Duration of the action potential and refractory period in fast fibers is shorter than slow fibers | ج   |       |
| d. Initial depolarization occurs in fast fibers with a rapid influx of $\text{Ca}^{2+}$ ions         | د   |       |
| Guyton p 126                                                                                         |     | منبع: |

|                                                                                            |       |    |
|--------------------------------------------------------------------------------------------|-------|----|
| Physiology                                                                                 |       | 28 |
| Regarding the Cardiac sympathetic nerve stimulation all the followings are correct EXCEPT: |       |    |
| a- Has positive inotropic effect.                                                          | الف   |    |
| b- Causes L-type $Ca^{2+}$ channel to spend more time in the open state.                   | ب     |    |
| c- Accelerates relaxation and shortens systole.                                            | ج     |    |
| d- Lengthens the AV nodal delay                                                            | د     |    |
|                                                                                            | منبع: |    |
| Guyton p: 128                                                                              |       |    |

|                                                                                          |       |    |
|------------------------------------------------------------------------------------------|-------|----|
| Physiology                                                                               |       | 29 |
| The ventricles are completely depolarized during which portion of the electrocardiogram? |       |    |
| a. PR interval                                                                           | الف   |    |
| b. QRS complex                                                                           | ب     |    |
| c. QT interval                                                                           | ج     |    |
| d. ST segment                                                                            | د     |    |
|                                                                                          | منبع: |    |
| Guyton p: 133                                                                            |       |    |

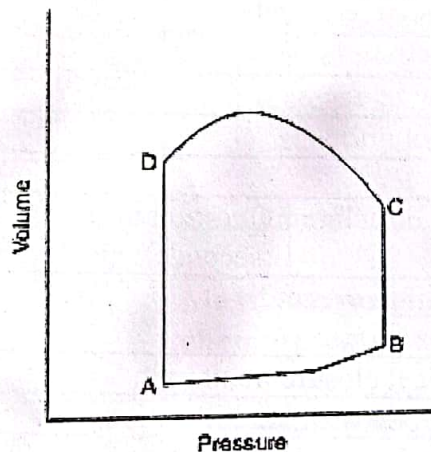
|                                                                                         |       |    |
|-----------------------------------------------------------------------------------------|-------|----|
| Physiology                                                                              |       | 30 |
| Which of the following is INCORRECT concerning diastolic depolarization at the SA node? |       |    |
| a. It results from an outward $K^{+}$ current, $iK$                                     | الف   |    |
| b. It results from the activation of $I_f$ carried mainly by Na funny channels.         | ب     |    |
| c. Its rate is decreased by sympathetic stimulation.                                    | ج     |    |
| d. It results from an inward $Ca^{++}$ current, $I_{Ca}$ .                              | د     |    |
|                                                                                         | منبع: |    |
| Guyton p: 124                                                                           |       |    |

|                                 |       |    |
|---------------------------------|-------|----|
| Physiology                      |       | 31 |
| What is the normal QT interval: | الف   |    |
| a. 0.03 seconds                 | ب     |    |
| b. 0.13 seconds                 | ج     |    |
| c. 0.16 seconds                 | د     |    |
| d. 0.35 seconds                 |       |    |
|                                 | منبع: |    |
| Guyton p: 133                   |       |    |



## Physiology

A 66-year old male presents with angina and dyspnea on exertion. Auscultation of the chest reveals a loud systolic murmur. He is diagnosed with aortic valve sclerosis. This stenosis of the aortic valve will cause a murmur that can be heard loudest at which interval shown



- between A and B
- between B and C
- between C and D
- between D and A

الف

ب

ج

د

منبع:

Guyton p: 118

## Physiology

Regarding the ECG findings of myocardial infarction (MI) all the following are correct EXCEPT:

33

- Elevation of the ST segment in recent (acute) MI.
- Deep and wide Q wave in old MI.
- Low voltage in massive MI .
- Prominent U wave.

الف

ب

ج

د

منبع:

Guyton p: 147

Physiology: Dr. Karimian

|    |                                                                                   |                               |  |
|----|-----------------------------------------------------------------------------------|-------------------------------|--|
|    | Blood circulation                                                                 | Question time<br>(seconds): → |  |
| 34 | Which of the following <u>is not</u> the homeostatic action of blood circulation? |                               |  |
|    | 1                                                                                 | Temperature regulation        |  |
|    | 2                                                                                 | Blood pressure                |  |
|    | 3                                                                                 | Cells Immunity                |  |
|    | 4                                                                                 | Feeding the cells             |  |

|    |                                                            |                                                                               |  |
|----|------------------------------------------------------------|-------------------------------------------------------------------------------|--|
|    | Blood circulation                                          | Question time<br>(seconds): →                                                 |  |
| 35 | In blood flow issue, which sentence <u>is not</u> correct? |                                                                               |  |
|    | 1                                                          | Reduction in RBC No. reduces critical closure point unit.                     |  |
|    | 2                                                          | Decrease in filling pressure will happen with vasodilatation.                 |  |
|    | 3                                                          | With increase in the artery diameter the pressure to the wall would increase. |  |
|    | 4                                                          | An increase in blood flow, the viscosity effect will decrease                 |  |

|    |                                                                    |                                                                            |  |
|----|--------------------------------------------------------------------|----------------------------------------------------------------------------|--|
|    | Blood circulation                                                  | Question time<br>(seconds): →                                              |  |
| 36 | In comparing between 2 kinds of circulation, which one is correct? |                                                                            |  |
|    | 1                                                                  | The blood volume in 2 circulation are equal.                               |  |
|    | 2                                                                  | The blood pressure in 2 circulation are equal.                             |  |
|    | 3                                                                  | The ejection fraction in big circulation is higher than small circulation. |  |
|    | 4                                                                  | The stroke volume in 2 circulations are equal                              |  |

|    |                                                   |                                                        |  |
|----|---------------------------------------------------|--------------------------------------------------------|--|
|    | Blood circulation                                 | Question time<br>(seconds): →                          |  |
| 37 | In the blood flow issue, select correct sentence: |                                                        |  |
|    | 1                                                 | Blood flow have direct relation with vessels length    |  |
|    | 2                                                 | Blood velocity have indirect effect with vessel length |  |
|    | 3                                                 | Resistance have indirect relation with vessels radius  |  |
|    | 4                                                 | Blood flow have direct relation with viscosity         |  |



|    |                                                                                                              |                               |  |
|----|--------------------------------------------------------------------------------------------------------------|-------------------------------|--|
|    | Blood circulation                                                                                            | Question time<br>(seconds): → |  |
| 38 | In the changing of the tendency flow from laminar to turbulent, which of the following has positive effects? |                               |  |
|    | 1                                                                                                            | Increase in viscosity         |  |
|    | 2                                                                                                            | Vessel partial occlusion      |  |
|    | 3                                                                                                            | Increase in velocity          |  |
|    | 4                                                                                                            | Reduction in diameter         |  |

|    |                                                                                                  |                                                              |  |
|----|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--|
|    | Blood circulation                                                                                | Question time<br>(seconds): →                                |  |
| 39 | Mean arterial flow in a person is 70 mm/ Hg how are the baroreceptor and chemoreceptor activity? |                                                              |  |
|    | 1                                                                                                | They don't have any activity                                 |  |
|    | 2                                                                                                | Baroreceptors are active and chemoreceptor inactive          |  |
|    | 3                                                                                                | only chemoreceptors are active and baroreceptor are inactive |  |
|    | 4                                                                                                | Both receptors are smoothly active                           |  |

|    |                                                                                                                                                                                                                                               |                               |  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--|
|    | Blood circulation                                                                                                                                                                                                                             | Question time<br>(seconds): → |  |
| 40 | Write the wanted amount and units of followings, respectively:<br><u>Right ventricle diastolic pressure</u> , <u>right atrium systolic pressure</u> , <u>Normal capillary tension</u> in mm/Hg and <u>pulse pressure</u> (from left to right) |                               |  |
|    | 1                                                                                                                                                                                                                                             | 80-25-60-90                   |  |
|    | 2                                                                                                                                                                                                                                             | 8-8-17-40                     |  |
|    | 3                                                                                                                                                                                                                                             | 25-20-17-40                   |  |
|    | 4                                                                                                                                                                                                                                             | 95-80-8-15                    |  |

|    |                                             |                                                |  |
|----|---------------------------------------------|------------------------------------------------|--|
|    | Blood circulation                           | Question time<br>(seconds): →                  |  |
| 41 | Which of the following causes tissue edema? |                                                |  |
|    | 1                                           | Reduction in capillary hydrostatic pressure    |  |
|    | 2                                           | Increase in capillary colloid osmotic pressure |  |
|    | 3                                           | Increase in filtration co-efficient            |  |
|    | 4                                           | Increase in lymphatic flow                     |  |

|    |                          |                                                                      |  |
|----|--------------------------|----------------------------------------------------------------------|--|
|    | Blood circulation        | Question time<br>(seconds): →                                        |  |
| 42 | select correct sentence. |                                                                      |  |
|    | 1                        | Increase in compliance Cause an increase in systolic blood pressure. |  |
|    | 2                        | Decrease in compliance Cause reduction in pulse pressure             |  |
|    | 3                        | Increase in blood volume increases pulse pressure                    |  |
|    | 4                        | Increase in stroke volume decreases systolic pressure.               |  |

|    |                                                                                                |                               |  |
|----|------------------------------------------------------------------------------------------------|-------------------------------|--|
|    | Blood circulation                                                                              | Question time<br>(seconds): → |  |
| 43 | Which of the following organs have highest and lowest blood volume in the moment respectively? |                               |  |
|    | 1                                                                                              | Heart- pulmonary organ        |  |
|    | 2                                                                                              | Muscle- CNS                   |  |
|    | 3                                                                                              | GI- thyroid                   |  |
|    | 4                                                                                              | Liver- skin                   |  |
|    | Source                                                                                         |                               |  |
|    | Description                                                                                    |                               |  |

|    |                                                       |                               |  |
|----|-------------------------------------------------------|-------------------------------|--|
|    | Blood circulation                                     | Question time<br>(seconds): → |  |
| 44 | Which of the following causes an increase in the PRU? |                               |  |
|    | 1                                                     | Anemia                        |  |
|    | 2                                                     | Heavy pain                    |  |
|    | 3                                                     | Organ amputation              |  |
|    | 4                                                     | Shock                         |  |

|    |                                                                       |                               |  |
|----|-----------------------------------------------------------------------|-------------------------------|--|
|    | Blood circulation                                                     | Question time<br>(seconds): → |  |
| 45 | Which of the following <u>does not</u> happen with histamine release? |                               |  |
|    | 1                                                                     | Edema                         |  |
|    | 2                                                                     | Allergic signs                |  |
|    | 3                                                                     | Arteries contraction          |  |
|    | 4                                                                     | Local vaso dilatation         |  |



|    |                                                                      |                               |
|----|----------------------------------------------------------------------|-------------------------------|
|    | Blood circulation                                                    | Question time<br>(seconds): → |
| 46 | Which of the following tissue has the least sympathetic innervation? |                               |
|    | 1                                                                    | Brain                         |
|    | 2                                                                    | Kidney                        |
|    | 3                                                                    | Intestine                     |
|    | 4                                                                    | Skin                          |

|    |                                                                                       |                               |
|----|---------------------------------------------------------------------------------------|-------------------------------|
|    | Blood circulation                                                                     | Question time<br>(seconds): → |
| 47 | In the normal heart working, which of the following is mostly used to produce energy? |                               |
|    | 1                                                                                     | Adenosine                     |
|    | 2                                                                                     | Glucose                       |
|    | 3                                                                                     | Galactose                     |
|    | 4                                                                                     | Fatty acids                   |

|    |                              |                                |
|----|------------------------------|--------------------------------|
|    | Blood circulation            | Question time<br>(seconds): →  |
| 48 | Active hyperemia happens in: |                                |
|    | 1                            | short term occlusion of artery |
|    | 2                            | increase in tissue metabolism  |
|    | 3                            | injury of tissue               |
|    | 4                            | increase in blood pressure     |

|    |                                                                                     |                                                                       |
|----|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
|    | Blood circulation                                                                   | Question time<br>(seconds): →                                         |
| 49 | In a constant resistant blood vessel, which of the following has higher blood flow? |                                                                       |
|    | 1                                                                                   | Pressure in the beginning 100 and in the end of vessel is 40 (mm/Hg)  |
|    | 2                                                                                   | Pressure in the beginning 250 and in the end of vessel is 200 (mm/Hg) |
|    | 3                                                                                   | Pressure in the beginning 80 and in the end of vessel is 10 (mm/Hg)   |
|    | 4                                                                                   | Pressure in the beginning 300 and in the end of vessel is 260 (mm/Hg) |

|    |                                                                                                                         |                               |  |
|----|-------------------------------------------------------------------------------------------------------------------------|-------------------------------|--|
|    | Blood circulation                                                                                                       | Question time<br>(seconds): → |  |
| 50 | Even with high changes in blood pressure, the tissue blood flow and supply do not change significantly. Why? Because of |                               |  |
|    | 1                                                                                                                       | sympathetic activity          |  |
|    | 2                                                                                                                       | para sympathetic activity     |  |
|    | 3                                                                                                                       | hormonal activity             |  |
|    | 4                                                                                                                       | auto regulation.              |  |

|    |                                                                                                                                                              |                               |  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--|
|    | Blood circulation                                                                                                                                            | Question time<br>(seconds): → |  |
| 51 | In a blood vessel, flow is 80 ml/sec and pressure in main and pulmonary circulations are 120 and 20 respectively. What is the PRU in these two circulations? |                               |  |
|    | 1                                                                                                                                                            | 1.5-0.25                      |  |
|    | 2                                                                                                                                                            | 16-25                         |  |
|    | 3                                                                                                                                                            | 4-0.8                         |  |
|    | 4                                                                                                                                                            | 8-12                          |  |

|    |                                                        |                                                                    |  |
|----|--------------------------------------------------------|--------------------------------------------------------------------|--|
|    | Blood circulation                                      | Question time<br>(seconds): →                                      |  |
| 52 | Which of the following sentence <u>is not</u> correct? |                                                                    |  |
|    | 1                                                      | Amputation causes increase in PRU                                  |  |
|    | 2                                                      | In a parallel system increase in vesseles decreases resistance     |  |
|    | 3                                                      | By developing the new vessels blood flow will increase.            |  |
|    | 4                                                      | Inhibition of sympathetic system increases critical closure point. |  |

|    |                                                  |                                                                                  |  |
|----|--------------------------------------------------|----------------------------------------------------------------------------------|--|
|    | Blood circulation                                | Question time<br>(seconds): →                                                    |  |
| 53 | In the veins pressure which sentence is correct? |                                                                                  |  |
|    | 1                                                | Pressure of the left atrium is the same of the central vein pressure.            |  |
|    | 2                                                | By increasing of pumping of right heart pressure of right atrium will increases. |  |
|    | 3                                                | Arteriole relaxation cause blood flow from arterioles to veins.                  |  |
|    | 4                                                | In the upright position the neck veins pressure is normally 4-6 mm/hg            |  |



|    |                                                               |                                                                                |  |
|----|---------------------------------------------------------------|--------------------------------------------------------------------------------|--|
|    | Blood circulation                                             | Question time (seconds): →                                                     |  |
| 54 | In the capillary filtration issue, which sentence is correct? |                                                                                |  |
|    | 1                                                             | Activation of pre- capillary sphincter is responsible for capillary vasomotion |  |
|    | 2                                                             | Proteins easily pass through to inter cellular space via capillary fenestra    |  |
|    | 3                                                             | Activation of lymphatic pump increases local edema.                            |  |
|    | 4                                                             | Protein content of lymph is equal with the plasma level of protein             |  |

|    |                                               |                                                                                             |  |
|----|-----------------------------------------------|---------------------------------------------------------------------------------------------|--|
|    | Blood circulation                             | Question time (seconds): →                                                                  |  |
| 55 | Which of the following <u>is not</u> correct. |                                                                                             |  |
|    | 1                                             | Atrial and pulmonary reflexes are sensitive to blood volume and regulates arterial pressure |  |
|    | 2                                             | In the bodies different positions, baroreceptors regulate blood pressure                    |  |
|    | 3                                             | Baroreceptors have an important role in long term regulation of blood pressure              |  |
|    | 4                                             | Activation of chemo receptors Cause reduction in blood pressure.                            |  |

|    |                                                                                 |                            |  |
|----|---------------------------------------------------------------------------------|----------------------------|--|
|    | Blood circulation                                                               | Question time (seconds): → |  |
| 56 | In which of the following vessels, turbulent flow never <u>does not</u> appear? |                            |  |
|    | 1                                                                               | Aorta artery               |  |
|    | 2                                                                               | Pulmonary artery           |  |
|    | 3                                                                               | Brachial artery            |  |
|    | 4                                                                               | Arterioles                 |  |

|    |                                                   |                                            |  |
|----|---------------------------------------------------|--------------------------------------------|--|
|    | Blood circulation                                 | Question time (seconds): →                 |  |
| 57 | Which of the following causes myogenic phenomena? |                                            |  |
|    | 1                                                 | Local regulation of tissue blood pressure. |  |
|    | 2                                                 | Local regulation of tissue blood flow      |  |
|    | 3                                                 | Reactive hyperemia                         |  |
|    | 4                                                 | Active hyperemia                           |  |

|    |                                                                                            |                                                                                     |  |
|----|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--|
|    | Blood circulation                                                                          | Question time<br>(seconds): →                                                       |  |
| 58 | Regarding the central regulatory system (vasomotor), which sentence <u>is not</u> correct? |                                                                                     |  |
|    | 1                                                                                          | C1 has direct vasoconstrictor effect on blood flow.                                 |  |
|    | 2                                                                                          | A1 has direct vasodilator effect on blood flow.                                     |  |
|    | 3                                                                                          | A2 receives informative neurons from peripheral.                                    |  |
|    | 4                                                                                          | Chemoreceptors have stimulatory effect on vasomotor and increase the blood pressure |  |

anatomy: Dr.Pasbakhsh

|    |                                                                                        |                               |  |
|----|----------------------------------------------------------------------------------------|-------------------------------|--|
|    | Anatomy of cardiovascular block                                                        | Question time<br>(seconds): → |  |
| 59 | Which of the following muscles is in the superficial layer of the erector spin muscle? |                               |  |
|    | 1                                                                                      | Longissimus thoracis          |  |
|    | 2                                                                                      | Multifidus                    |  |
|    | 3                                                                                      | Rotatores                     |  |
|    | 4                                                                                      | Levator costarom              |  |

|    |                                                                   |                               |  |
|----|-------------------------------------------------------------------|-------------------------------|--|
|    | Anatomy of cardiovascular block                                   | Question time<br>(seconds): → |  |
| 60 | Which of the following structures is more anterior than the rest? |                               |  |
|    | 1                                                                 | Right ventericle              |  |
|    | 2                                                                 | Right Vagus nerve             |  |
|    | 3                                                                 | Left atrium                   |  |
|    | 4                                                                 | Right atrium                  |  |

|    |                                                                           |                               |  |
|----|---------------------------------------------------------------------------|-------------------------------|--|
|    | Place the name of the course or test                                      | Question time<br>(seconds): → |  |
| 61 | The right margin of heart is formed mostly by which champer of the heart? |                               |  |
|    | 1                                                                         | Left atrium                   |  |
|    | 2                                                                         | Right atrium                  |  |
|    | 3                                                                         | Left ventricle                |  |
|    | 4                                                                         | Right ventricle               |  |



|    | Place the name of the course or test                                          | Question time (seconds): → |
|----|-------------------------------------------------------------------------------|----------------------------|
| 62 | Which of the following statement is correct?                                  |                            |
| 1  | 4 <sup>th</sup> Anterior intercostal artery arises from Musculophrenic artery |                            |
| 2  | Thoracic aorta pass posterior to left atrium                                  |                            |
| 3  | Subcostal artery arises from internal thoracic artery                         |                            |
| 4  | Left and right internal thoracic veins drains to hemi Azygous vein            |                            |

|    | Place the name of the course or test                              | Question time (seconds): → |
|----|-------------------------------------------------------------------|----------------------------|
| 63 | Which of the following is visible anterior to the thoracic Aorta? |                            |
| 1  | Accessory hemiazygous vein                                        |                            |
| 2  | Pericard                                                          |                            |
| 3  | Hemi Azygous vein                                                 |                            |
| 4  | Thoracic vertebra                                                 |                            |

|    | Place the name of the course or test                                                       | Question time (seconds): → |
|----|--------------------------------------------------------------------------------------------|----------------------------|
| 64 | In the case of the intervertebral disc, all of the following statements are correct except |                            |
| 1  | Annulus fibrosis are located in the peripheral part of the disc                            |                            |
| 2  | Disc herniation usually occurs in the postero lateral direction                            |                            |
| 3  | Posterior portion of Annulus fibrosis is thicker than anterior part                        |                            |
| 4  | Intervertebral disc is composed of the nucleus pulposus and Annulus fibrosis               |                            |

|    |                                                                           |
|----|---------------------------------------------------------------------------|
| 65 | Which of the following structures is going to innervate by phrenic nerve? |
| 1  | Pectoralis major                                                          |
| 2  | Myocard of heart                                                          |
| 3  | Trapezius muscle                                                          |
| 4  | Fibrous pericard                                                          |

|    | Place the name of the course or test                                         | Question time<br>(seconds) → |
|----|------------------------------------------------------------------------------|------------------------------|
| 66 | Which of the following structure is found inferior to the arch of the aorta? |                              |
| 1  | Right recurrent laryngeal nerve                                              |                              |
| 2  | Left vagus nerve                                                             |                              |
| 3  | Deep cardiac plexus                                                          |                              |
| 4  | Superficial cardiac plexus                                                   |                              |

|    | Place the name of the course or test                                                                                                                                                       | Question time<br>(seconds) → |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 67 | During a heart procedure the surgeon accidentally damages the vein that is accompanied by the anterior interventricular artery, which of the following veins is most likely to be damaged? |                              |
| 1  | Great cardiac vein                                                                                                                                                                         |                              |
| 2  | Small cardiac vein                                                                                                                                                                         |                              |
| 3  | Middle cardiac vein                                                                                                                                                                        |                              |
| 4  | Left marginal vein                                                                                                                                                                         |                              |

|    | Place the name of the course or test                              | Question time<br>(seconds) → |
|----|-------------------------------------------------------------------|------------------------------|
| 68 | Which of following ribs has a single articular facet on its head? |                              |
| 1  | Third rib                                                         |                              |
| 2  | second rib                                                        |                              |
| 3  | Tenth rib                                                         |                              |
| 4  | Ninth rib                                                         |                              |

|    | Place the name of the course or test                                                                           | Question time<br>(seconds) → |
|----|----------------------------------------------------------------------------------------------------------------|------------------------------|
| 69 | Pathological mass is find in superior mediastinum, which of the following structures <u>may</u> be compressed? |                              |
| 1  | Left phrenic nerve                                                                                             |                              |
| 2  | Hemi Azygous vein                                                                                              |                              |
| 3  | Right recurrent laryngeal nerve                                                                                |                              |
| 4  | Thoracic Aorta                                                                                                 |                              |



|      | Place the name of the course or test                                  | Question time (seconds): → |
|------|-----------------------------------------------------------------------|----------------------------|
| ✓ 70 | The first rib is going to articulate with all of the following except |                            |
|      | 1 Manobrium of sternum                                                |                            |
|      | 2 Body of the first thoracic vertebra                                 |                            |
|      | 3 Transvers process of first thoracic vertebra                        |                            |
|      | 4 Disc between first and second vertebra                              |                            |

|    |                                                                 |
|----|-----------------------------------------------------------------|
| 71 | All of the following artery arises from thoracic Aorta except ? |
|    | 1 5 <sup>th</sup> left and right posterior intercostal artery   |
|    | 2 Superior phrenic                                              |
|    | 3 Subcostal                                                     |
|    | 4 Common carotid                                                |

|      | Place the name of the course or test | Question time (seconds): → |
|------|--------------------------------------|----------------------------|
| ✓ 72 | Where is crista terminalis?          |                            |
|      | 1 In right atrium                    |                            |
|      | 2 In left atrium                     |                            |
|      | 3 In right ventricle                 |                            |
|      | 4 In left ventricle                  |                            |

|      | Place the name of the course or test | Question time (seconds): → |
|------|--------------------------------------|----------------------------|
| ✓ 73 | Where the Thoracic duct has seen?    |                            |
|      | 1 In posterior mediastinum           |                            |
|      | 2 Anterior to bifurcation of Aorta   |                            |
|      | 3 Anterior to arch of Aorta          |                            |
|      | 4 In middle mediastinum              |                            |

|      | Place the name of the course or test               | Question time (seconds): → |
|------|----------------------------------------------------|----------------------------|
| ✓ 74 | Where we should listen to Aortic valve sounds?     |                            |
|      | 1 In left 5th intercostal space over apex of heart |                            |
|      | 2 In 6 <sup>th</sup> right intercostal space       |                            |
|      | 3 In 2th left intercostal space                    |                            |
|      | 4 In 2th right intercostal space                   |                            |

|    | Place the name of the course or test                           | Question time (seconds): → |
|----|----------------------------------------------------------------|----------------------------|
| 75 | Which of following joints are the type of cartilaginous joint? |                            |
|    | 1                                                              | First Costo Vertebral      |
|    | 2                                                              | Sternal angle              |
|    | 3                                                              | First Costo Sternal        |
|    | 4                                                              | Second Costo transvers     |

|    | Place the name of the course or test                                              | Question time (seconds): → |
|----|-----------------------------------------------------------------------------------|----------------------------|
| 76 | The 5th posterior intercostal arteries on each side are derived from which artery |                            |
|    | 1                                                                                 | Subclavian                 |
|    | 2                                                                                 | Left Internal thoracic     |
|    | 3                                                                                 | Right internal thoracic    |
|    | 4                                                                                 | Thoracic aorta             |

|    | Place the name of the course or test                                                                          | Question time (seconds): →  |
|----|---------------------------------------------------------------------------------------------------------------|-----------------------------|
| 77 | All of the following muscles are innervated by the anterior branch (ventral) of the intercostal nerves except |                             |
|    | 1                                                                                                             | Erector spine               |
|    | 2                                                                                                             | Serratus posterior inferior |
|    | 3                                                                                                             | Serratus posterior superior |
|    | 4                                                                                                             | Inter costals               |

|    | Place the name of the course or test                                                 | Question time (seconds): →                                     |
|----|--------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 78 | In the case of the sympathetic chain, which of the following definitions is correct? |                                                                |
|    | 1                                                                                    | The sympathetic chain is also seen in the abdomen              |
|    | 2                                                                                    | Greater splanchnic nerves dose not contain myelin              |
|    | 3                                                                                    | Lesser splanchnic nerve dose not contain myelin                |
|    | 4                                                                                    | White rami is along the entire length of the sympathetic chain |



|    | Place the name of the course or test                                           | Question time (seconds): → |
|----|--------------------------------------------------------------------------------|----------------------------|
| 79 | Lymphatic drainage of all the following areas into the thoracic duct except?.. |                            |
| 1  | Right lower limb                                                               |                            |
| 2  | Left lower limb                                                                |                            |
| 3  | Right posterior second intercostal space                                       |                            |
| 4  | Left posterior first intercostal space                                         |                            |

|    | Place the name of the course or test                                  | Question time (seconds): → |
|----|-----------------------------------------------------------------------|----------------------------|
| 80 | Which of the following ligaments is connected to the costal tubercle? |                            |
| 1  | Lateral costo transvers                                               |                            |
| 2  | Superior costo transvers                                              |                            |
| 3  | Posterior longitudinal                                                |                            |
| 4  | Anterior longitudinal                                                 |                            |

|    | Place the name of the course or test                                        | Question time (seconds): → |
|----|-----------------------------------------------------------------------------|----------------------------|
| 81 | Which of the following options is correct ?                                 |                            |
| 1  | The thoracic aorta begins on the right side of the fourth thoracic vertebra |                            |
| 2  | The pericardio co phrenic vessels arises from thoracic Aorta                |                            |
| 3  | The thymus is found in the posterior mediastinum                            |                            |
| 4  | The left brachiocephalic vein is longer than the right                      |                            |

|    | Place the name of the course or test                                                     | Question time (seconds): → |
|----|------------------------------------------------------------------------------------------|----------------------------|
| 82 | All the following options are correct except....                                         |                            |
| 1  | The abdominal aorta divides into two branches around the fourth lumbar vertebra          |                            |
| 2  | In surface Anatomy position of Pulmonary valve is at the level 3th left costal cartilage |                            |
| 3  | In surface Anatomy position of Aortic valve is at the level left 3th intercostal space   |                            |
| 4  | The left posterior intercostal artery is longer than the right                           |                            |

|    | Place the name of the course or test                                 | Question time<br>(seconds): → |
|----|----------------------------------------------------------------------|-------------------------------|
| 83 | In orifice of the all following vesseles valve is visible except.... |                               |
|    | 1 Inferior vena cava                                                 |                               |
|    | 2 Pulmonary trunk                                                    |                               |
|    | 3 Superior vena cava                                                 |                               |
|    | 4 Coronary sinus                                                     |                               |

Histology: Dr. Rastgar

|    |                                             |
|----|---------------------------------------------|
| 84 | In arteriol:                                |
|    | 1 Tunica media contaion 20-30 muscle layers |
|    | 2 No internal elastic lamina                |
|    | 3 Elastic fibers are high amount            |
|    | 4 Tunica externa is thick                   |

|    |                                                                            |
|----|----------------------------------------------------------------------------|
| 85 | In the normal histological structure of the different types of capillaris: |
|    | 1 Sinusoids have large pores                                               |
|    | 2 Endothelial cell are continuous in fenestrations capillaris              |
|    | 3 Continuous capillaris have pores                                         |
|    | 4 Thick basement membranes are in Sinusoids                                |

|    |                                                           |
|----|-----------------------------------------------------------|
| 86 | All of statements are correct about thymus EXCEPT:        |
|    | 1 Medulla include continus capillary                      |
|    | 2 Hassal's body are remanence of Epithelioreticular cells |
|    | 3 Thymus luboles include Cortex and Medulla               |
|    | 4 Hassal's body are basophil structure in Medulla         |

|    |                                                          |
|----|----------------------------------------------------------|
| 87 | Which of these are described lymphatic tissue?           |
|    | 1 In Diffuse lymphatic tissue capsule is absent          |
|    | 2 lymphatic nodes found in clusters at connective tissue |
|    | 3 Lymphatic nodules include Cortex and Medulla           |
|    | 4 Spleen have just oval-shaped lymphstic masses          |

|    |                                                        |
|----|--------------------------------------------------------|
| 88 | Which of lymphatic tissue are Primary lymphoid organs? |
|    | 1 Thymus- Spleen                                       |
|    | 2 lymphatic nodes- Tonsil                              |
|    | 3 Bone marrow-thymus                                   |
|    | 4 Lymphatic nodules- lymphatic nodes                   |

|    |                                                    |
|----|----------------------------------------------------|
| 89 | In lymphatic nodes.....:                           |
| 1  | Subcapsular sinus was seen in medulla              |
| 2  | Medullary cords extend from the Medulla            |
| 3  | Paracortex have no B lymphocyte                    |
| 4  | Medullary sinuses extended inward from the capsule |

|    |                                                       |
|----|-------------------------------------------------------|
| 90 | All of these statements are correct in Spleen EXCEPT: |
| 1  | Central artery are in white pulp.                     |
| 2  | Red pulp located in medulla.                          |
| 3  | Billrout's cord are clusters of lymphocytes.          |
| 4  | Stave cells are in open circulatory                   |

|    |                                      |
|----|--------------------------------------|
| 91 | Muscular artery include.....:        |
| 1  | Elastic fibers in Intima             |
| 2  | Thick tunica externa or adventitia   |
| 3  | Tunica media have 4 layers of muscle |
| 4  | Internal & External Elastic Lamina   |

|    |                                          |
|----|------------------------------------------|
| 92 | Which of these antibodies are in saliva? |
| 1  | IgE                                      |
| 2  | IgA                                      |
| 3  | IgM                                      |
| 4  | IgD                                      |

|    |                                                   |
|----|---------------------------------------------------|
| 93 | In the normal histological structure of arttries: |
| 1  | Elastic artry include more muscles cells          |
| 2  | Aorta is Muscular artry                           |
| 3  | Intima include squamos epithelial cells           |
| 4  | Arteries have tight junctions                     |

Embryology: Dr. Amidi

|    |                                                                                                                                                                                                                  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 94 | For items 1-3, select the <b>one</b> lettered option from the following list that is most closely associated with each numbered item below. Options in the list may be used once, more than once, or not at all. |
|    | shunts blood from the right atrium to the left atrium                                                                                                                                                            |
|    | 1 foramen ovale                                                                                                                                                                                                  |
|    | 2 ductus arteriosus                                                                                                                                                                                              |
|    | 3 ductus venosus                                                                                                                                                                                                 |
|    | 4 tricuspid valve                                                                                                                                                                                                |

|    |                                                                |
|----|----------------------------------------------------------------|
| 95 | shunts blood from the umbilical vein to the inferior vena cava |
|    | 1 foramen ovale                                                |
|    | 2 ductus arteriosus                                            |
|    | 3 ductus venosus                                               |
|    | 4 tricuspid valve                                              |

|    |                                                             |
|----|-------------------------------------------------------------|
| 96 | derived in part from <u>atrioventricular cushion tissue</u> |
|    | 1 foramen ovale                                             |
|    | 2 ductus arteriosus                                         |
|    | 3 ductus venosus                                            |
|    | 4 tricuspid valve                                           |

|    |                                                                            |
|----|----------------------------------------------------------------------------|
| 97 | During vascular development, the hepatic portal venous system arises from: |
|    | 1 posterior cardinal veins.                                                |
|    | 2 umbilical veins.                                                         |
|    | 3 vitelline veins.                                                         |
|    | 4 subcardinal veins.                                                       |

|    |                                    |
|----|------------------------------------|
| 98 | The 3rd Aortic Arch gives rise to: |
|    | 1 Proximal internal carotid        |
|    | 2 Right subclavian                 |
|    | 3 Pulmonary artery                 |
|    | 4 Common carotid                   |



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|    |                     |                          |
|----|---------------------|--------------------------|
| 99 | supracardinal veins |                          |
|    | 1                   | Give rise to SVC         |
|    | 2                   | Give rise to IVC         |
|    | 3                   | Give rise to Portal vein |
|    | 4                   | Give rise to Azygos vein |

|     |                                                                        |                             |
|-----|------------------------------------------------------------------------|-----------------------------|
| 100 | Which of the following is not associated with the Tetralogy of Fallot? |                             |
|     | 1                                                                      | Large VSD                   |
|     | 2                                                                      | Hypertrophy right ventricle |
|     | 3                                                                      | Overriding aorta            |
|     | 4                                                                      | Aortic estenosis            |